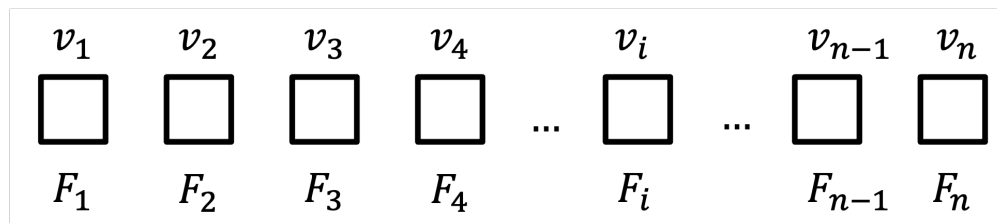


## Prophet Inequalities

You're at a casino gambling, and are offered to play the following game. Items will arrive one-by-one. As an item arrives, you see its value. You may only take a single item, and once you take an item, the game ends. A priori, you know the *distribution* of each item. At some point there will be a red item with the red distribution of values, and at some point there will be a blue item with the blue distribution of values, and so forth. However, you do not know the order of items (it is adversarial), and you do not know the exact values of the items (they are drawn from their specific distributions). Your goal is to come up with an algorithm that competes with the *prophet* who is all knowing, so knows the realization of values and the arrival order.

That is,  $n$  items will arrive in adversarial order. Item  $i$  (which is a label, not necessarily the order) has value  $v_i$  drawn from known distribution  $F_i$ . Your goal is to determine an algorithm ALG such that the value you get from gambling competes with the prophet who always gets  $\max_i v_i$ . However, your competition is over the randomness of the values that are drawn, so you only have to compete with  $\text{OPT} = \mathbb{E}_{\mathbf{v}}[\max_i v_i]$ . We also use ALG to denote the algorithm's expected value.

The values are independent and nonnegative, with finite expected maximum. The adversary chooses the order before seeing the values. If we take no item, we get zero.



To summarize:

- Goal: Pick one item; maximize its value.
- Gambler knows distribution for each item.
- Order is adversarial.
- Inspect each item online (see  $v_i$ ) and irrevocably decide whether to take or pass forever.
- Compete with  $\text{OPT} = \mathbb{E}_{\mathbf{v}}[\max_i v_i]$ .

**Example.** The first item has value 1. The second has value 10 with probability  $1/5$ , and 0 otherwise. What does the prophet get in expectation? What can the gambler get?

## Choosing a threshold

We will use a *threshold algorithm*: choose a number  $T \geq 0$  before any items arrive, and take the first item whose value is at least  $T$ . If no item meets the threshold, we get zero.

How should we choose  $T$ ?

**Attempt 1.** Suppose we choose  $T = 1$  in the following example:

$$v_1 = 1, \quad v_2 = \begin{cases} 100 & \text{with probability } 1/10, \\ 0 & \text{otherwise.} \end{cases}$$

The item of value 1 arrives first. What does our algorithm get in expectation? What does the prophet get? How bad can the comparison become if we increase 100 to a much larger value?

**Attempt 2.** What if we set  $T = \text{OPT}$ , the prophet's expected value? Consider the same arrival order, but make the large value much rarer:

$$v_1 = 1, \quad v_2 = \begin{cases} 100 & \text{with probability } 1/1000, \\ 0 & \text{otherwise.} \end{cases}$$

What threshold does this give, and what does our algorithm get? Compare with the prophet. What happens if the probability of 100 gets even smaller?

What goes wrong when the threshold is too low? Too high?

## The one-half guarantee

Prove the following.

**Theorem 1.** *There is a threshold algorithm ALG such that when the gambler takes the first item whose value is at least  $T$ ,  $\text{ALG} \geq \frac{1}{2}\text{OPT}$ .*

Determine what threshold  $T$  to use and prove this statement using the following steps:

1. Divide what the algorithm yields from an item (in expectation) into exactly the threshold and the surplus above the threshold.
2. Lower bound your surplus term.
3. Set your threshold in order to combine like-terms and have OPT pop out.

*Proof.* Let  $p$  denote the probability that *some* (at least one)  $v_i \geq T$ . For the proof, relabel the items in arrival order. Write  $(v_i - T)^+ = \max\{v_i - T, 0\}$  for the surplus above  $T$ .

## Posted prices

We reinterpret our problem as follows: we have a single item to sell, and  $n$  buyers arrive online with their values for the item drawn from distributions. We price the item at  $T$ . Observe that the first term above is the *expected revenue* earned from selling the item to the buyers at  $T$ , and the second term is the *expected utility*, or surplus. Welfare is equal to revenue + utility.

Each buyer has one chance to buy at price  $T$  while the item remains. They cannot come back later. We sell to the first buyer who accepts, and assume a buyer accepts when their value equals the price. The seller has no value for keeping the item.

Why does a buyer want to accept exactly when their value is at least  $T$ ?

**Example.** Return to our first example, now with two buyers. The first buyer has value 1. The second has value 10 with probability  $1/5$ , and 0 otherwise. What price does our theorem tell us to use?

Work out what happens at this price:

| $v_2$ | Probability | Buyer served | Welfare | Revenue | Total utility |
|-------|-------------|--------------|---------|---------|---------------|
| 0     | $4/5$       |              |         |         |               |
| 10    | $1/5$       |              |         |         |               |

What are the expected welfare, revenue, and utility? Check that welfare equals revenue + utility and is at least  $\frac{1}{2}\text{OPT}$ .

What if we instead set the price to 1? How do welfare and revenue change?

## The connection to virtual values

We have used the prophet inequality to design a mechanism that approximates the optimal social welfare. Could we use it to approximate the optimal expected revenue?

For this question, suppose buyers' values are independent and their distributions are continuous and regular, as in L4. Recall that regularity means each buyer's virtual value is nondecreasing in their value, and expected revenue equals expected virtual welfare for a truthful mechanism with zero payments at value zero.

What should we apply the prophet inequality to, and what prices should we offer?